ZHAO Xuecen (Jack)

Robotics Systems Engineer

Robot software, simulation and LiDAR safety | Python, ROS 2, Unreal Engine 5, Jetson Nano, Raspberry Pi, ESP32

Hong Kong (Permanent Resident)

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PROFESSIONAL SUMMARY

Robotics Systems Engineer (BEng Mechanical Engineering, PolyU 2026) on Stairio, an award-winning stairwell inspection robot; sole developer of its Unreal Engine 5 software twin, running the robot's own V7 Python controller against virtual CAN motor, IMU and camera devices (**610**-test suite). Built the mechanics, AI and control of a ROS 2 robot car whose **deterministic LiDAR safety arbiter gates velocity commands**, including those from LLM tool calls.

AWARDS

- **Gold Medal with Congratulations of the Jury**, International Exhibition of Inventions of Geneva 2026 (Stairio, project team)
- **Hong Kong first runner-up**, James Dyson Award 2026, through to the international stage (Stairio, project team)
- **Most Popular Solution Award**, PolyU Faculty of Engineering Inter-departmental Final Year Project 2025/26 (Vandi, team)

CORE SKILLS

Hardware: ROS 2 and SLAM (SLAM Toolbox), LiDAR, IMU and cameras (RPLiDAR A1, dual-fisheye), Raspberry Pi and Jetson (Pi Zero 2 W, Jetson Nano), Motor and servo control, CAN (SocketCAN audit, virtual CAN), ESP32 firmware (C++), Kinematics and PID (5-DOF FK/IK), SolidWorks CAD

AI: Simulation and digital twins (Unreal Engine 5), YOLO training and TensorRT (YOLO11n on Jetson Nano), OpenCV, LLM tool calling, Vision-language models, Human-in-the-loop design

Software: Python: FastAPI and Flask, Automated testing (610 simulator tests), Live video streaming, Linux, Git and Docker, Firebase and Cloud Run

Communication: Technical writing and diagrams, Team leadership, English, Cantonese and Putonghua (IELTS 7.0)

EXPERIENCE

Robotics Systems Engineer

Mar 2026 - Present

Stairio Limited, Hong Kong

- **Software twin:** Sole developer of an Unreal Engine 5 twin in which the robot's own V7 controller (not a reimplementation) drives virtual CAN motor, IMU, environment and camera devices with fault injection in a 20-floor stairwell; **it passes all 8 acceptance-runner stages** over real HTTP, bounded by a five-level evidence plan.
- **Stream latency:** Measured the robot's live camera stream (Pi Zero 2 W, dual-fisheye, 5 fps target) in 5-minute runs: 421 ms median / 538 ms P95 local, 591 ms / 774 ms via a Cloud Run relay. Fixed a silent ~10.5 s cloud stall (the open connection hid it from socket errors) with publisher 2 s frame-ack and browser 5 s freshness reconnects; **a 90 s rerun had zero reconnects**.
- **Firmware audit:** Audited the live robot (MPU6500, BME688, CANable2/C610 on CAN) and pinned its firmware baseline with checksums; found the IMU sampled at 50 Hz but published at 10 Hz; **reproduced 10 firmware defects** as offline regression fixtures (e.g. a re-armed reversal dwell holding current at zero through a 5 s test).
- **IMU navigation:** Built an ideal scene-derived IMU model; replay showed the legacy floor counter missing a landing at 120 RPM as its 0.5 s gravity filter lags pitch changes. Wrote a versioned extension of the team's newer floor detector (gyro-predicted gravity, split-corner fix, >=2 s calibration before every Start) that **detects all 37 landings at 30 and 120 RPM** in replay.
- **Motion arbitration:** Built single-owner control for the Unreal console, web operator and cloud clients (session-checked commands, takeover never replays Start, 1,500 ms motion lease under a 3 s owner lease). Fixed background-tab drive stalls by moving 200 ms lease renewal into a server-side stream, proved by **a 279 s, 105 m continuous run at 120 RPM**.
- **Video pipeline:** Streamed live 360° video from simulated cameras at the CAD lens positions in the Pi's 1280×480 dual-fisheye format, so the unmodified production operator can dewarp it. Hemisphere-only sampling and cached lookup maps **cut first-call panorama projection time by 46%** (pixel-identical output), so full-resolution captures finish while both live views stream.
- **Behaviour design:** Designed the Stairio 360 on-robot behaviour: setup/routine state machine, position quality states (anchored/tracking/degraded/lost) with uncertainty, floor identity not from IMU alone, one supervisor gating all actuation.
- **Firmware and CAD:** Contributed to the Stairio S1's ESP32-P4 camera firmware (C/C++) and supported commissioning, capture review and field demonstrations; adapted parts of the team's SolidWorks robot CAD (enclosure, mounts, packaging), testing the changes on the physical robot.

- ***Also:*** Built the bilingual Stairio website and the Stairio HR hiring system, and made the launch video and Blender renders.

Independent Consulting and Design Engineer
Sep 2022 - Sep 2026

JZfreelance, Remote

- Ran an independent engineering practice for **295+** clients over four years (scoping to follow-up): SolidWorks drawings and parametric models, ANSYS design checks and 3D-printed prototypes, plus web apps and LLM workflow automation.

SELECTED PROJECTS

Vandi - Fuel-Cell-Powered Robot Car (Final Year Project)
2025 - 2026

- **Architecture:** My part of a three-department team: mechanical design, AI, simulation and control (teammates: fuel chemistry, PCB/BMS); one Jetson Nano script starts **8** processes (LiDAR, SLAM, chassis, odometry, Flask, UI).
- **Safety arbiter:** Put teleop, chat, LLM-tool and auto-drive velocity commands behind one deterministic LiDAR function, since detector, autonomy and LLM can all be wrong. It checks a 40–60° window in the direction of travel, stops at 0.10 m and inside 0.20 m scales linear speed to a 12% floor but angular only to 35% so it can turn away; it **refuses forward motion on scans older than 0.65 s**.
- **Motion control:** Built the ROS 2 bridge (/cmd_vel if subscribed, else serial to the Yahboom board, both behind the safety function) and single-owner motion control; **the LLM never writes motor values** (ten fixed tools, max two calls per turn), and a ten-pattern guardrail rejects unsafe spoken commands.
- **Sensor fusion:** Fused vision-model boxes with 2D LiDAR scans to answer 'how far away is the chair?' **with no depth camera**: bearing = (0.5 - box centre) × 70° HFOV; range = nearest return in a box width × 70° × 0.85 window (6–18°), scan <=1.5 s old.
- **Perception:** Trained a three-class YOLO11n sign detector on the Jetson Nano (640 px, 80 epochs, on-device TensorRT export; **0.989** mAP50 on a same-collection validation split, not field accuracy); built the HSV traffic-light colour stage and a 60 s LiDAR-scored free-drive mode that re-scans eight directions every 0.25 s.

Car Spoiler Operations - Active Spoiler Prototype
2026

- Turned an adjustable spoiler prototype into a five-stage mechatronic system (**the only full marks in the ME42011 class**): a six-link linkage on a geared positional servo, driven from the grounded-side link, under staged positional control.
- Wrote ESP32-C3 firmware (Arduino C++, PlatformIO) in which HTTP handlers only set a target and a 20 ms loop slews the servo 1° per tick inside a hard **20°–160°** clamp. One press moves one stage, Stop holds position, and the servo centres at boot.

Other Projects

- **Robot Basketball (2025):** Team leader of **8**, **first place** in the EEE/ME inter-departmental contest; shooter design in SolidWorks.
- **Auto Shooter (2023):** Raspberry Pi robot; OpenCV HSV targeting drives PCA9685 pan/tilt servos; PID/PI tracking tuning.
- **5-DOF Robot Arm (2022):** CAD design; FK/IK and workspace study in simulation, then tested on the physical arm.

ADDITIONAL EXPERIENCE

Full Stack Developer
Jan 2026 - Feb 2026

Feelings AI, Hong Kong

- Built physics, camera modes and socket-server multiplayer for an AI 3D scene builder; integrated Gemini/Ollama LLM agents.
- ***Also:*** Designed the SQLite/Knex scene database (initialisation, migration, seeding scripts) and an undo/redo history.

AI Features Developer Intern
Dec 2025 - Jan 2026

Inspire Education Limited, Hong Kong

- Built AI personalised-learning features for the FunMA maths platform, plus Swift iOS app features.

On-site Student Assistant, *CERN, Geneva, Switzerland*: science tour guide explaining LHC operations — May 2019 - Jun 2019

Digital Engineering Intern, *VineUp (acquired by Graduway), Gloucester, UK*: helped migrate data — Apr 2019 - May 2019

Technical Intern, *North Oxford BMW, Oxford, UK*: engine diagnostics and performance testing — Oct 2018 - Nov 2018

EDUCATION

BEng (Hons) Mechanical Engineering
Sep 2022 - Jul 2026

The Hong Kong Polytechnic University

- Final year project grade A. Electives: Perceptual Robotics, Fundamentals of Robotics, Artificial Intelligence in Products.

British A-Levels and GCSE
2018 - 2022

Oxford International College, Oxford, UK. A-Levels: Mathematics, Further Mathematics, Physics, Computer Science.

References: Prof. Li-Ta HSU, Prof. AN Liang and Prof. ZHANG Dan (PolyU); contact details on request.